

Laboratory 1

Design of a Combinational Logic Circuit

INTRODUCTION: This laboratory will demonstrate the basic design principles of a combinational logic circuit for a specified application. Students will also validate the circuit design by building the circuit and verifying the Truth Table.

EDUCATIONAL OBJECTIVES:

1. Design a combinational circuit and implement it with multiplexers.
2. Simulate the circuit using Proteus.
3. Validate the circuit design by building the circuit on a breadboard.

EXPERIMENTAL OBJECTIVES:

1. Design a parity generator.
2. Design a vote counter.

PRE-LAB:

Simulate the Even Parity Generator using Proteus.

COMPONENT NEEDED:

- | | |
|--|---|
| 1. 74151 8x1 Multiplexer IC | 1 |
| 2. 7447 BCD-to-7-Segment Decoder IC | 1 |
| 3. 7-Segment LED Display | 1 |
| 4. Other Logic Gates (NOT, 3-input AND gates and etc.) | |
| 5. Bread Board | |

BACKGROUND

A. MULTIPLEXER

Multiplexer is for transmitting a large number of information units over a smaller number of channels or lines. A digital multiplexer is a combinational circuit that selects binary information from one of many input lines and directs it to a single output line. The selection of a particular input line is controlled by a set of selection lines. Normally there are (2^n) input line and (n) selection lines whose bit combination determine which input is selected.

IC Description - 74151 MUX

74151 is an 8 line-to-1 line multiplexer. It has the schematic representation shown in Figure 1. Selection lines S_2 , S_1 and S_0 select the particular input to be multiplexed and applied to the output.

Strobe (**S**) acts as an enable signal. If (**S = 1**), the chip 74151 is disabled and the output (**Y = 0**). If (**S = 0**) then the chip 74151 is enabled and it functions as a multiplexer. Table 1 and Figure 1 show the functionality of multiplex 74151 in terms of select lines (**S₀**, **S₁** and **S₂**).

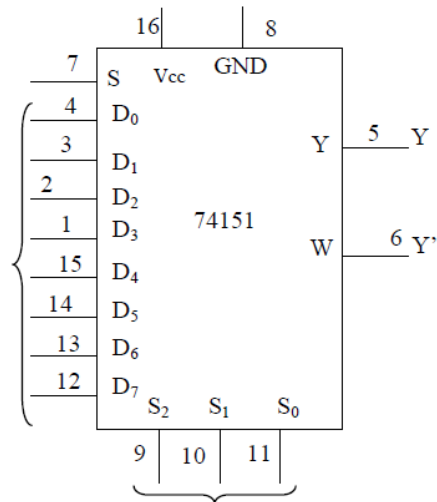


Figure 1: IC type 74151 8×1 Multiplexer

Table 1 : Functionality of multiplex 74151

Strobe	Select Lines			Output
S	S ₂	S ₁	S ₀	Y
1	x	x	x	0
0	0	0	0	D ₀
0	0	0	1	D ₁
0	0	1	0	D ₂
0	0	1	1	D ₃
0	1	0	0	D ₄
0	1	0	1	D ₅
0	1	1	0	D ₆
0	1	1	1	D ₇

B. DECODER

A decoder is a combinational circuit that converts binary information from (***n***) input lines to a maximum of (***2ⁿ***) output lines.

IC Description - 7447 Decoder

A7-Segment LED display contains 7 LEDs. Each LED is called a segment and they are identified as (**a, b, c, d, e, f** and **g**) segments in Figure 3.

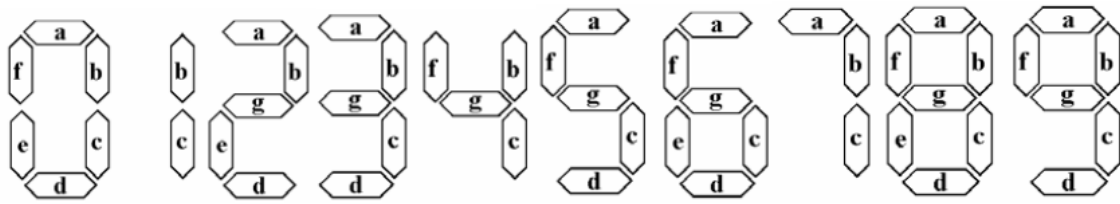


Figure 3: Digits represented by 7-segments

The display has 7 inputs, each connected to an LED segment. All anodes of LEDs are tied together and joined to 5 V (Common Anode type). A limiting resistance network must be used at the inputs to protect the 7- Segment from overloading. BCD inputs are converted into 7-Segment inputs using a decoder as shown in Figure 4.

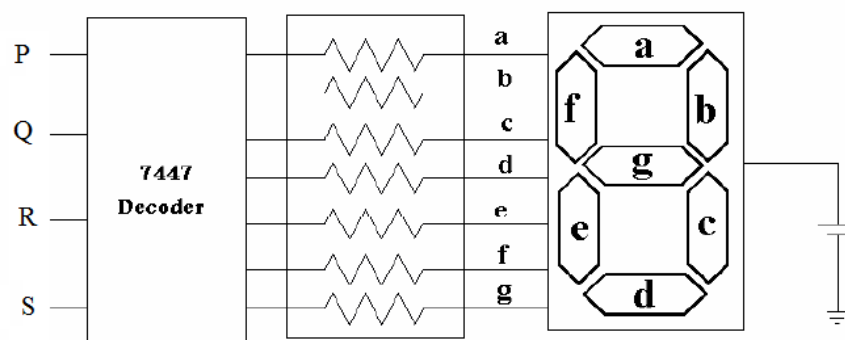


Figure 4: BCD inputs are converted into 7-Segment inputs using a decoder

The input to the decoder is a binary coded decimal (BCD) code and the outputs of the systems are the seven segments (a, b, c, d, e, f and g). Pin connections are shown in Figure 5. For further information and pin connections, see the specification sheet for Decoder and 7-Segment unit.

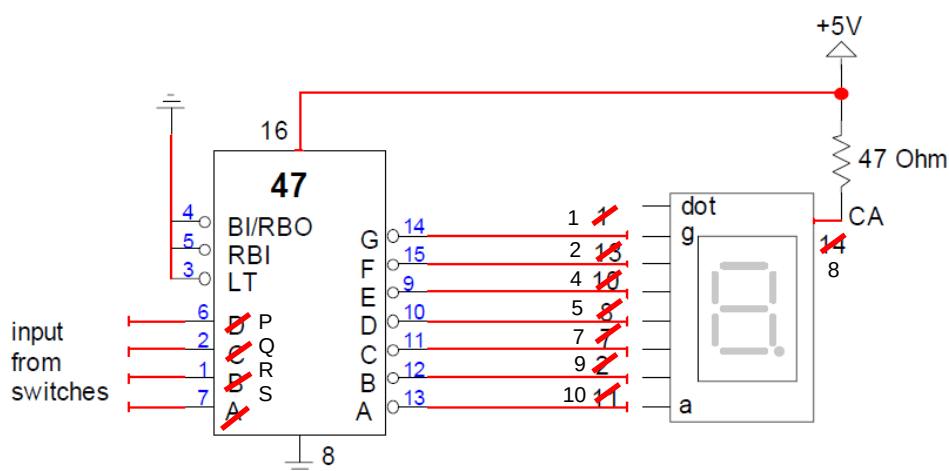


Figure 5: BCD to 7-Segment Decoder and 7-Segment Display

Note: It is needed to design a combinational circuit which would simulate the decoder function for each segment of the 7-Segment Display.

SAFETY PRECAUTIONS

1. Always turn off power to the circuit when changing the circuit.
2. Only reapply power after verifying that the circuit is properly wired with the relevant voltage values.

DESIGN PROBLEM

A small corporation has 9 shares of stock. The Corporation Board consists of 4 members (Chairman, two senior members and one junior member). Each member of the Corporation Board has a voting right at a Corporation Financial meeting. The committee has agreed to assign the share(s) for the vote of each member.

Chairman (C)	: 4 shares
Senior Member 1 (M_1)	: 2 shares
Senior Member 2 (M_2)	: 2 shares
Junior Member (J)	: 1 share

The committee wants to automate the voting process so that the voting machine that gets the inputs from each member and then displays the total number of shares. Each member has switch to close when voting **YES (Logic 1)** and to open when voting **NO (Logic 0)** for his/her vote.

Suppose that you are a Design Engineer in the same corporation and have been asked to design such a voting machine. The only ICs available to you are one 8x1 MUX, 3-input AND gates, one 7-Segment Decoder, one 7-Segment LED display and one XOR gate.

DESIGN PROCEDURE

1. Draw a simplified block diagram of such a voting machine (No need to make the exact logic diagram).
2. There are four inputs to this machine (one from each member) and the output is a 4-bit binary number (**C**, **M_1** , **M_2** and **J**). Draw the truth table for the output function.

[illegible]

- Table 3: The truth table for the inputs and the output of the 7-Segment Display Unit.

[illegible]

6. Show your complete circuit in working condition after completing the logic diagram for whole voting machine.
7. Implement it in hardware on a bread board.

FINAL DESIGN:

Your laboratory instructor will inform you when the design is due.

UNIVERSITY OF RUHUNA

FACULTY OF ENGINEERING

DEPARTMENT OF ELECTRICAL AND INFORMATION ENGINEERING

EE4302 – Digital Electronics

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Laboratory Report

			Signature of the Instructor
Student Number			
Laboratory Conducted Date			
Laboratory Report	Due Date		
	Submitted Date		

Evaluation (Instructor Use Only)			
	Evaluation Component	Allocated Marks Out of 10.0	Student Achieved Marks
1	Pre-Lab Computing (Parity Generator)	2.5	
2	Design (Voting Machine)	-	
	Phase I - Multiplexer Design	5.0	
	Phase II- 7-Segment Display Design	2.5	
	Total	10.0	
Signature of the Evaluator			

Pre-Lab Computing :

1. Simulate an even parity generator using Proteus.

Problem: Design an Even Parity Generator using a 74151 multiplexer

Parity is an extra bit inserted into a code to make sure that the code has been received correctly. Even parity bit means that the number of ones (1s) in the code including the parity bit is an even number.

- a) Fill the output column of the truth table in Table 1 for a 5-bit code in which four bits (**A**, **B**, **C** and **D**) represents the information to be sent and the fifth bit (**X**) represents the parity bit.
- b) The inputs **B**, **C** and **D** correspond to the select inputs of 74151. Complete the truth table in Table 1 by filling in the last column with **0**, **1**, **A** or $\bar{A}$.
- c) Simulate the circuit using Proteus. Use 74151 multiplexer, binary switches for inputs and binary probes for outputs in the simulation. Simulate the circuit and test each input combination filling in the Table 1.

Table 1: Simulate the circuit and test each input combination

Inputs				Output	Connect Data to
A	B	C	D	X	

Note: You should complete the simulation prior to the Laboratory session and show it to laboratory Instructor. Also you should be able to answer the question raised by the laboratory Instructor.